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Role

Content Video Generation, Audio Scripting, FMNPG, LinkedIn Automation & Feedback Loop

Tech Stack

FastAPI Gemini / Groq LLM pool gTTS FFmpeg LinkedIn UGC API (OAuth 2.0) SQLite
Closed-loop feedback memory

1. Content Video Generation

- Built the video output path of the **Media Decision Engine (Stage 2)**: turns a brand brief (Jade / DoctorShield / Jaguar Transit) into a multi-scene video — one visual per scene, assembled into a final MP4.
- Handles the visual-source decision per scene: cloud AI provider (Gemini/Hugging Face) when available and fast, falling back to a local GPU model (Stable Diffusion 1.5) when the cloud provider is unavailable, and finally a branded placeholder card if no AI source can produce the image — each case is labeled with the correct “Visual source” badge so nothing is misrepresented as AI-made when it wasn’t.

2. Audio Scripting

- Built the script-to-speech layer: takes the Content Agent’s generated script/hooks/copy and produces narration text, converts it via **gTTS**, and syncs it with burned-in captions via **FFmpeg**.

3. FMNPG

- Worked on FMNPG. *(description pending — fill in what this stands for and what it does before committing.)*

4. LinkedIn Automation — “The Hands” (Stage 5)

- Built the full **LinkedIn Publisher**: authenticated UGC Post dispatch (video or image binary) to the authorized LinkedIn profile via **OAuth 2.0**, using `urn:li:person` identifiers.
- Built the **Connect LinkedIn** flow (redirect to LinkedIn's own login/consent screen, scoped to posting on the user's own profile only — never a company page, never anything beyond posting).
- Built the **dispatch record** system: every publish attempt (real or simulated) is logged with outcome, post ID (for real posts), and error detail if it failed — status never claims success that didn't happen.
- Built the **background worker** ("Trigger Worker Pass") that publishes everything currently approved-and-ready in one batch.
- Implemented the **Autonomous Approval** gate (Stage 4): system-level `approval_status="AUTO_APPROVED"`, `approved_by="SYSTEM"`, `human_approved=false`, with full audit provenance — this is what lets content flow straight from compliance-pass to publish-ready without a manual click, per the project's decision to auto-resolve human review.

5. Feedback Loop (Closed-Loop Memory)

- Built the lessons-learned system: every reject/edit reason from Review Center is captured as a permanent lesson tied to a brand/topic.
- These lessons are injected as few-shot guidance into the Content Agent on future runs, so the same mistake isn't repeated.
- Built the **Closed-Loop Memory** screen: lists every learned lesson, what it applies to, how many times it's been reinforced, with the ability to deactivate a lesson that's no longer relevant.

How the full loop works: Content Agent (Stage 1) drafts copy + applies past lessons → Media Decision Engine (Stage 2) generates video/audio → Compliance Gate (Stage 3) scores against the 12-rule MAS/MOH/PDPA rubric, auto-correcting up to 3 times → Autonomous Approval (Stage 4) marks it publish-ready → LinkedIn Publisher (Stage 5) dispatches it and logs the result → any human edit/reject anywhere in this chain writes a new lesson back into Stage 1 for the next run.

Notes

Fill in what **FMNPG** stands for and its function, plus specific file paths (e.g.

`content_agent.py` , `linkedin_publisher.py` , `feedback_store.py`) **before committing to GitHub.**